

**ANALYSIS OF PLASMA POSITION IN THAILAND
TOKAMAK-1 USING TOROIDAL FILAMENT MODEL**

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entitled

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USING TOROIDAL FILAMENT MODEL**

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“Young opportunity sparked hopeful ideas, nurturing expansion.” We started off as a small research group of three individuals which later spanned out much over my expectation. From a small group consisting of a bachelor student and two supervisors, to a group spanning over many institutes including Thailand Institute of Nuclear Technology. I would like to express my indescribable gratitude to all the collaborators that made this project possible and profound thanks to my project advisors Associate Professor Apiwat Wisitsorasak and Mr. Suebsak Suksaengpanomrung who provided me this great opportunity.

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ABSTRACT

Nuclear fusion is a promising future energy source, offering clean and virtually limitless power by replicating the nuclear reactions that occur in stars. To maintain stability within a tokamak, a negative-feedback control system is essential. This senior project presents two methods for determining the plasma position inside a circular tokamak. The first approach, the *Toroidal Filament Model*, estimates the plasma position based on the magnetic field generated by the plasma. The second method, known as *Optical Boundary Reconstruction* (commonly referred to as *OFIT*), analyzes captured plasma images to reconstruct the plasma boundary. Both models are applied to experimental data from Thailand Tokamak-1. While the results from the two methods show similar trends in plasma position—indicating the models' potential for experimental application—exact quantitative agreement is not achieved due to limitations such as magnetic signal noise and uncertainties in camera calibration parameters.

KEY WORDS : PLASMA COLUMN POSITION, POLOIDAL PLANE,
TOROIDAL FILAMENT MODEL, OFIT,
THAILAND TOKAMAK-1

50 pages

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Energy production is essential for modern civilization; however, fossil fuels cause pollution, and renewable sources face challenges such as inconsistency and site dependency. To overcome these limitations and provide low-emission, nuclear fusion offers a promising alternative. It shares advantages with nuclear fission—such as high energy output and environmental adaptability—while avoiding its drawbacks, including long-lived radioactive waste, meltdown risks, and reliance on radioactive fuel. One of the devices designed to achieve nuclear fusion is a torus-shaped machine called a tokamak.

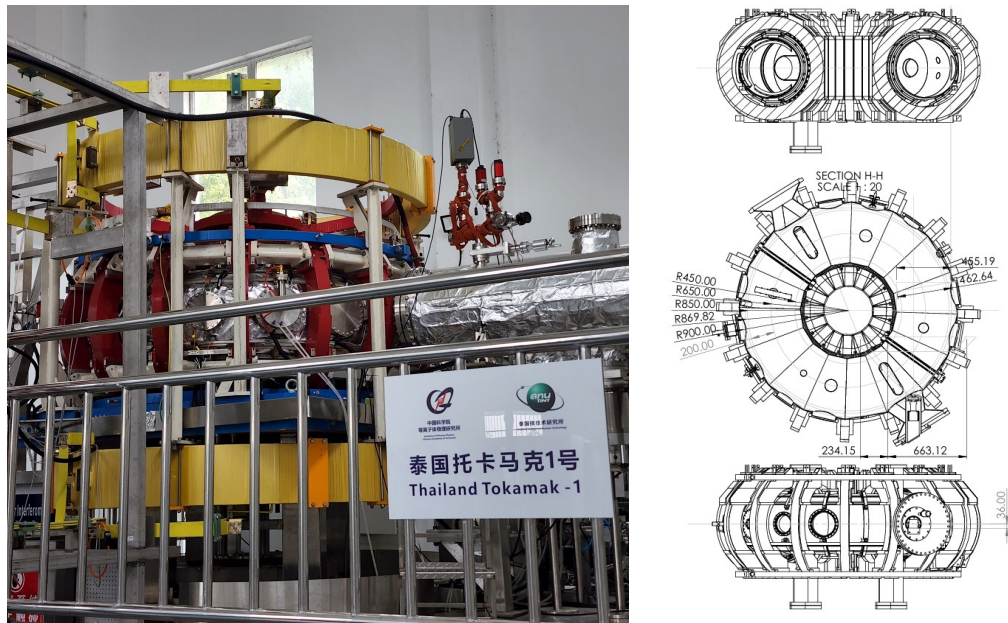


Figure 1.1: Photo and diagram of TT-1 tokamak

Thailand Tokamak-1 (TT-1), previously known as the HT-6M tokamak (Figure 1.1), is a small-sized device with a circular cross-section. Its parameters are listed in Figure 1.2. Two major magnetic fields are present in TT-1: the toroidal field and the

Vacuum Chamber	
Major radius (R_0)	0.65 m
Minor radius (R_m)	0.2 m
Average radius to each magnetic probes (r)	0.321 m
Toroidal Field (TF)	16 coils
Toroidal field	4.52 Tesla
Max. Ripple	6.53%
Turn current	7.64 kA
Flat top time	100 ms
Poloidal Field (PF)	7 coils
Central Solenoid coils	1 piece
Ohmic Heating coils	4 coils
Vertical Field	2 coils
Plasma volume control	Limiter
Density	$10^{18} - 10^{19}$
Electron temperature (eV)	~ 500

Figure 1.2: Parameters of TT-1 tokamak [1]

poloidal field. The toroidal field, which runs along the torus, is generated by magnetic coils encircling the device. What distinguishes the tokamak is its poloidal field system [2]. In addition to the poloidal field produced by the poloidal field coils (circular coils located at the top and bottom of the chamber), a poloidal magnetic field is also generated by the plasma current induced through a central transformer. The combination of toroidal and poloidal fields produces a helical magnetic field that helps to stabilize the plasma.

Various diagnostic instruments are installed in TT-1 to study different plasma characteristics. These include magnetic sensors, flux loops, saddle loops, CCD diagnostics, HCN laser interferometers, H-alpha detectors, and hard X-ray (HXR) detectors [1]. Despite this array of diagnostics, plasma position determination within the tokamak has not yet been pursued.

Accurate measurement of the plasma position within TT-1 is essential for stable plasma confinement. Figure 1.3 shows an image of confined plasma within the device. The confinement is controlled by magnetic fields from various coils surrounding the tokamak. However, instabilities may occur during confinement, posing a major challenge—small perturbations can grow rapidly and lead to large-scale instability [2]. The

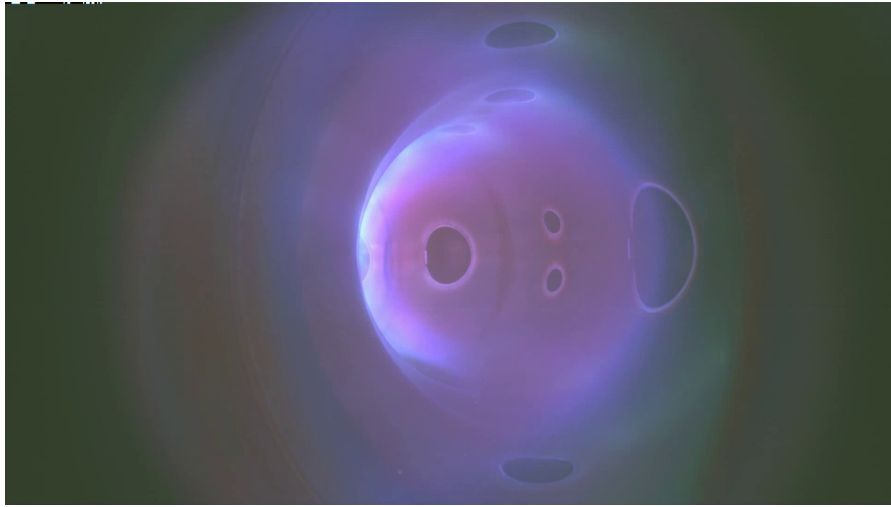


Figure 1.3: Image captured by a CCD camera showing confined plasma and the interior of TT-1

plasma may shift toward the vessel wall, contaminating it with impurities or damaging the tokamak's structure. Therefore, knowledge of the plasma position is crucial for providing real-time feedback control to mitigate displacement. Moreover, plasma position data can be used for further simulations and analyses, such as evaluating the plasma's magnetic field distribution.

1.2 Objectives

This project aims to develop two methods for determining the plasma position in the TT-1 tokamak. The main objectives are:

1. Derive the Toroidal Filament Model for the TT-1 tokamak.
2. Derive the Optical Boundary Reconstruction method (OFIT) for the TT-1 tokamak.
3. Cross-validate the derived methods using experimental data.

The first method, known as the Toroidal Filament Model, calculates the plasma centroid (or column) position based on the magnetic profile generated by the plasma. The second method, Optical Boundary Reconstruction (OFIT), uses plasma images. In this project, both models are adapted for the TT-1 tokamak: the Toroidal Filament Model is modified

to accommodate the TT-1's magnetic probe configuration, while OFIT is implemented with enhancements to reduce the effect of noise pixels resulting from high plasma optical emissions. These methods are applied to experimental data from TT-1 to estimate plasma position and to cross-validate their results.

1.3 Scopes of this work

This project focuses on methodologies for calculating the plasma position in TT-1 and is therefore applicable to tokamaks with similar characteristics. The scope of the Toroidal Filament Model is limited to tokamaks with a circular magnetic probe layout and measurable plasma-induced magnetic signals. The scope of OFIT is limited to plasmas with visible and distinguishable boundaries.

This work is an extension of the paper “*Calculation Methodologies for Plasma Position in TT-1 Tokamak*” [3]. In this thesis, detailed derivations and calculation procedures are provided. All Python-based programs developed and used in this project are available at: <https://github.com/P-Attapon/TT1PlasmaColumnPosition>.

CHAPTER 2

LITERATURE REVIEW

Plasma within tokamaks possesses many characteristics that can be analyzed to determine its column position. Some of the methods used to calculate the plasma column position are presented in this chapter.

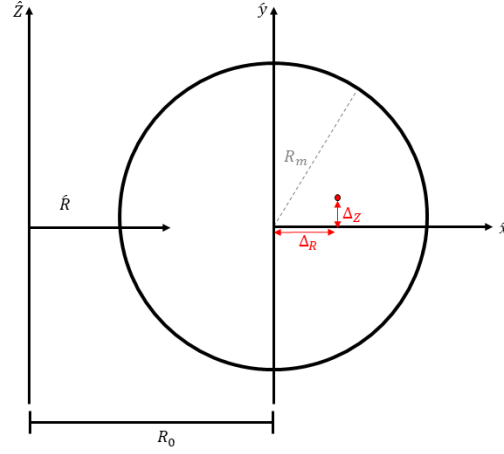


Figure 2.1: Poloidal plane of TT-1 showing with plasma column position

To define the problem of determining the plasma position—often referred to as the “column position”—relevant terminology must first be established. Figure 2.1 shows the cross-sectional view of the tokamak, known as the poloidal plane. On this plane, the plasma column position (Δ_R, Δ_Z) is defined as the displacement of the plasma’s centroid from the tokamak’s geometric center along the radial and vertical axes. In the figure, the major radius (R_0) is the distance from the tokamak center to the center of the torus shaped vessel, while the minor radius (R_m) corresponds to the radius of the circular cross-section of the vessel.

2.1 Toroidal Filament Model

The Toroidal Filament Model was originally proposed by Quin et al. [4] to calculate the plasma column position in configurations where magnetic probes are aligned to measure only the radial component of the magnetic field generated by circulating charges in the plasma. The magnetic field data recorded by these probes are used to estimate the plasma column position. This model approximates the plasma as a closed-loop current with a δ -function plasma density profile¹. The model relies on two assumptions: axial symmetry of the plasma along the toroidal direction and small displacement of the plasma column.

The Toroidal Filament Model estimates the plasma column position by establishing a fitted relationship between the actual displacement (Δ_R, Δ_Z) and respective first-order approximation (Δ_x, Δ_y) . The first order approximation corresponds to the scenario that the major radius of the tokamak approaches infinity. The model's strength lies in this fitting relationship, as it allows for pre-computation and lookup during runtime, enabling efficient computation.

Three coordinate systems are employed in this model (Figure 2.1):

- The tokamak cylindrical coordinate system (R, ϕ, Z) , with the origin at the tokamak center.
- A 2D Cartesian coordinate system (x, y) known as the poloidal plane, where $\hat{x} \parallel \hat{R}$ and $\hat{y} \parallel \hat{Z}$, with the origin at the center of the vessel.
- The polar coordinate system (r, θ) corresponding to the poloidal plane.

The transformation between the cylindrical and 2D coordinate systems involves the major radius:

$$\begin{aligned} R &= R_0 + x, & Z &= y, \\ x &= r \cos(\theta), & y &= r \sin(\theta). \end{aligned}$$

¹Here, the δ -function density refers to modeling the plasma as a current-carrying wire, similar to those found in elementary electrodynamics.

Further extensions of the model—including plasma minor radius calculation and the incorporation of a quadratic plasma density profile—are explored in [4].

In Quin et al.'s work, the column position $(\Delta_{\parallel}, \Delta_{\perp})$ at time step t_n is computed as a function of the first-order displacement approximation (Δ_x, Δ_y) and power series coefficients (α_i, β_i) , as follows:

$$\begin{aligned}\Delta_{\parallel}(t_n) &= \sum_{i=0}^3 \beta_i(t_{n-1}) \cdot D_x^i(t_n), \quad n = 1, 2, 3, \dots, \\ \Delta_{\perp}(t_n) &= \sum_{i=0}^3 \alpha_i(t_{n-1}) \cdot D_y^i(t_n), \quad n = 1, 2, 3, \dots,\end{aligned}\tag{2.1}$$

where

$$D_x = \frac{D^2}{2A} \frac{V_3 - V_1}{V_3 + V_1}, \quad D_y = D \frac{V_2 - V_4}{V_2 + V_4}.\tag{2.2}$$

Here, V_i represents the magnetic signal measured by probe i . In the original implementation for the TJ-1 tokamak (which has a rectangular cross-section), D and A represent the distances from the poloidal plane center to the horizontal and vertical edges, respectively.

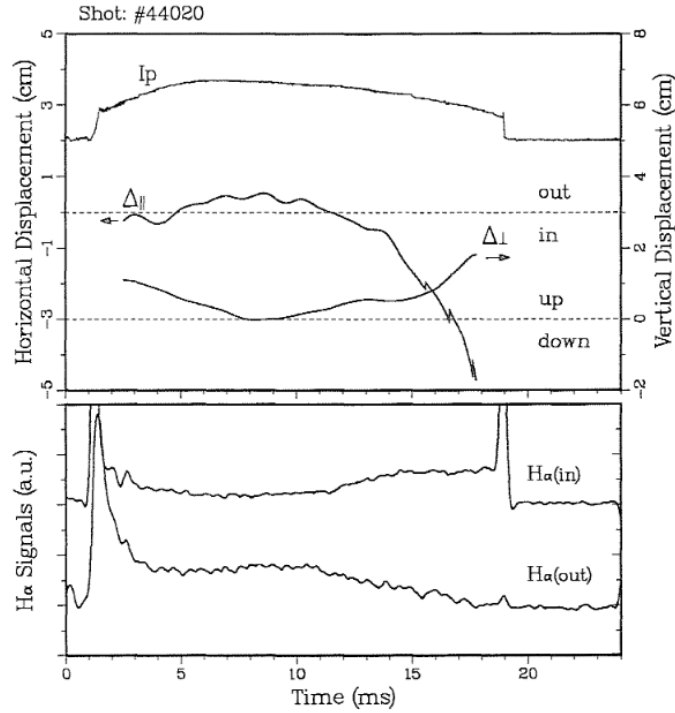


Figure 2.2: Comparison of plasma column position with two H_{α} signals during a discharge in TJ-1 tokamak [5]

Figure 2.2 presents results from the Toroidal Filament Model, alongside the plasma current I_p and H_α emission data. $H_\alpha(\text{in})$ and $H_\alpha(\text{out})$ correspond to signals from the inner and outer sides of the vessel, respectively. These signals allow verification of the radial plasma shift ($\Delta||$), as a displacement toward the inner wall increases $H_\alpha(\text{in})$ and decreases $H_\alpha(\text{out})$, and vice versa. At approximately 11 ms, the plasma begins shifting inward, corresponding to a rise in $H_\alpha(\text{in})$ and a drop in $H_\alpha(\text{out})$ [5].

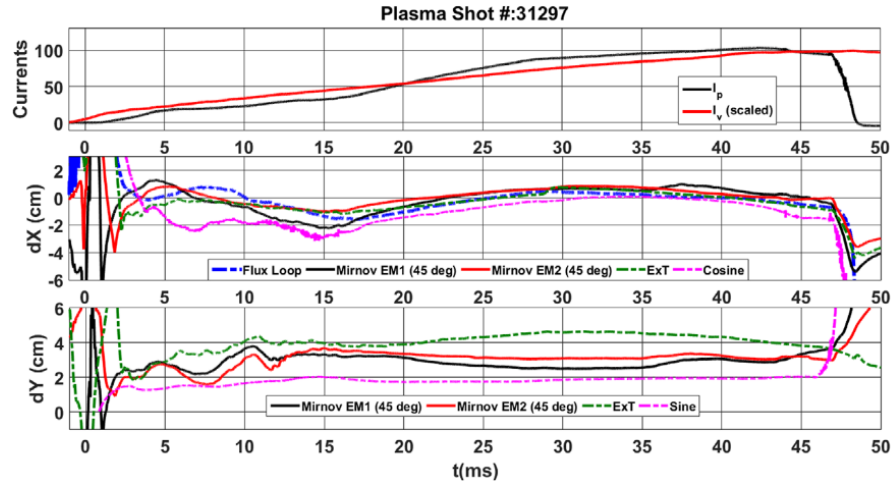


Figure 2.3: Implementation of the Toroidal Filament Model in the ADITYA tokamak. The top panel shows plasma current; middle and bottom panels show results of the Toroidal Filament Model and other diagnostics [6].

Figure 2.3 shows the application of the Toroidal Filament Model to the ADITYA tokamak, following correction of the magnetic signals. The model's results (Mirnov EM1 and EM2 at 45°) are compared with diagnostics from flux loops, cosine coils, and external tangential magnetic probes (ExT) [6]. Good agreement is observed from different diagnostic systems. From the image, at approximately 5 ms, the plasma stabilizes near the tokamak center. A temporary inward shift occurs at 10 ms, followed by a dramatic upward and inward displacement at 47 ms.

2.2 Optical Boundary Reconstruction (OFIT)

The Optical Boundary Reconstruction method (OFIT), introduced by Hommen et al. [7], is a diagnostic technique used in tokamaks such as MAST (UK) and EAST (China). OFIT captures electromagnetic radiation in the optical spectrum—primarily

bremsstrahlung—from decelerating plasma charges, using a high-speed CCD camera. The recorded image is then used to reconstruct the plasma boundary and estimate the column position.

OFIT models the plasma as a thin toroidally symmetric emissive surface. Due to the camera's nature, each pixel records line-integrated light intensity along a specific line of sight. Therefore, all emission along that sightline contributes to the pixel's brightness. OFIT assumes that edge pixels, corresponding to sightlines tangent to the plasma surface, exhibit peak intensity relative to their neighbors. These edge pixels are then projected onto the poloidal plane using a pinhole camera model.

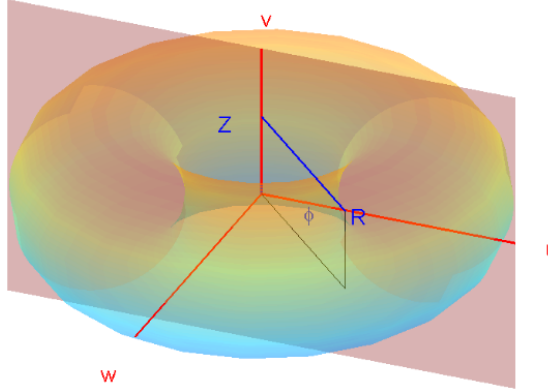


Figure 2.4: Tokamak coordinate system with projection plane [7]

Figure 2.4 visualizes the coordinate systems used in OFIT. Edge pixels detected in the image are located in pixel coordinates (x_e, y_e) and must be transformed to poloidal coordinates (R_e, Z_e) . However, reconstructing a 3D surface from a single 2D camera image is inherently infeasible; thus, OFIT assumes toroidal symmetry². OFIT employs three coordinate systems: pixel coordinates (x, y) , projection plane coordinates (u, v) —shown as the pink plane in Figure 2.4—and poloidal plane coordinates (R, Z) . The transformation occurs in three main steps:

1. **Plasma Edge Detection:** An edge detection algorithm is applied to the plasma image (e.g., Figure 1.3) to extract the plasma boundary in pixel coordinates (x_e, y_e) .

²Deviations from perfect symmetry—such as ripple effects—can later be incorporated [7].

2. **Transformation to Projection Plane:** The edge points are transformed from pixel coordinates to projection plane coordinates (u_e, v_e) using the camera's intrinsic and extrinsic calibration parameters (see Subsection B), which define the camera's position, orientation, focal length, and internal parameters.
3. **Transformation to Poloidal Plane:** Assuming toroidal symmetry, each edge point's azimuthal angle ϕ_e in tokamak cylindrical coordinate can be computed and used to transform edge points into poloidal coordinates (R_e, Z_e) using:

$$\begin{aligned}\tan(\phi_e) &= \frac{(u_e - u_c) - \frac{du_e}{dv_e}(v_e - v_c)}{w_c}, \\ R_e &= \frac{u_e w_c}{w_c \cos(\phi_e) + (u_e - u_c) \sin(\phi_e)}, \\ Z_e &= v_c + (v_e - v_c) \frac{w_c - R_e \sin(\phi_e)}{w_c}.\end{aligned}\tag{2.3}$$

Here, (u_c, v_c, w_c) denotes the camera position. The term $\frac{du_e}{dv_e}$ describes the curvature of the plasma edge in the projection plane and requires numerical evaluation [7].

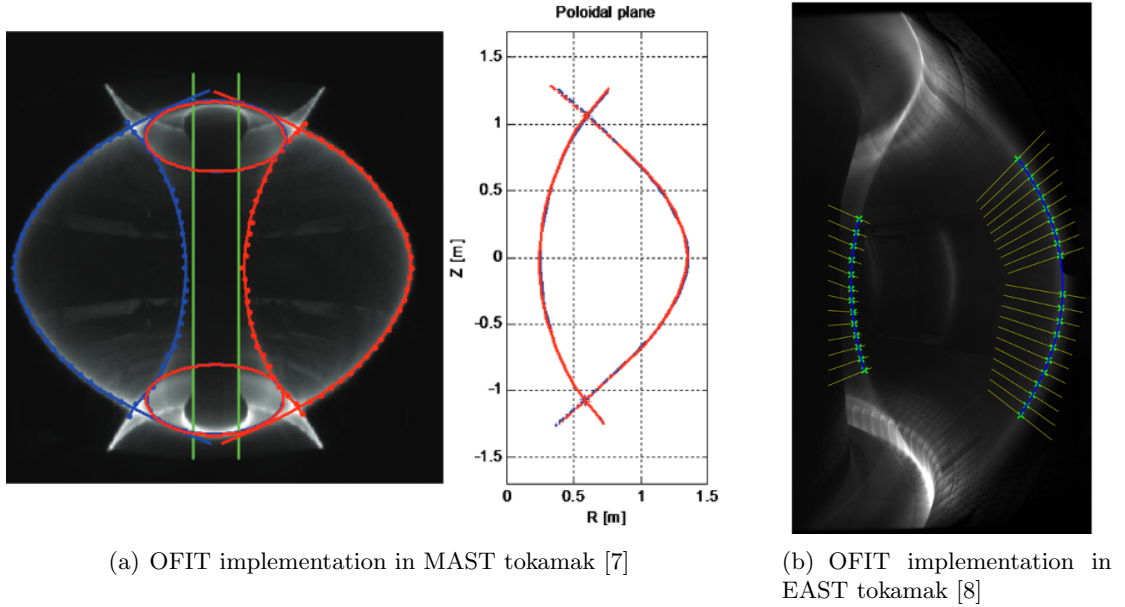


Figure 2.5: OFIT implementation in MAST and EAST tokamaks

Though the coordinate transformation framework is similar across implementations, the edge detection algorithm may vary. The method used in this project closely follows that used in MAST [7], where pixels with row-maxima intensity within a region

of interest (ROI) are identified as the plasma edge (Figure 2.5(a)). In EAST [8], pixels with highest intensity along predefined line segments are used (Figure 2.5(b)).

In the original work by Hommen et al. [7], OFIT was applied to MAST, where the CCD setup captured both plasma halves. Results from each half were compared against EFIT. Agreement between the two OFIT halves and EFIT was observed, with offsets of approximately 14 mm between result obtained from OFIT and EFIT.

Chen et al. [8] enhanced this method by introducing feature line segments for edge detection and real-time camera calibration to reduce uncertainty due to camera motion and allowing real-time execution.

2.3 Grad-Shafranov Equation (EFIT)

Although not implemented in this project, another approach to calculate plasma column position is solving the Grad-Shafranov (GSS) equation [9], commonly known as EFIT. This method reconstructs detailed magnetic flux surfaces based on magnetohydrodynamic (MHD) properties, including the plasma pressure and current profiles. Although highly accurate, EFIT requires a significantly more complex numerical solution.

The Grad-Shafranov equation is a nonlinear partial differential equation:

$$\Delta^* \Psi + \mu_0 R^2 \frac{dp}{d\Psi} + g \frac{dg}{d\Psi} = 0. \quad (2.4)$$

where Ψ is the poloidal magnetic flux, p is the plasma pressure, and g is the toroidal field function. Given boundary conditions, solving this equation yields the poloidal flux profile, from which the magnetic axis (interpreted as the plasma column position) can be identified. Figure 2.6 shows the reconstructed poloidal magnetic flux distribution of a D-shaped plasma, with the magnetic axis representing the column position.

Plasma Equilibrium

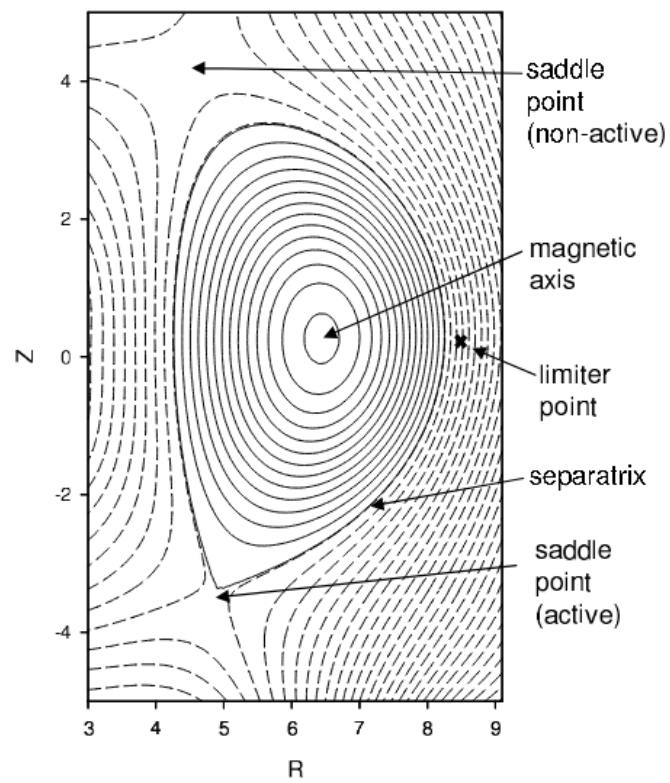


Figure 2.6: Example result of Ψ from the Grad-Shafranov equation [9]

CHAPTER 3

METHODOLOGY

3.1 Toroidal Filament Model for Plasma Position in TT-1

The Toroidal Filament Model proposed by Quin et al. [4] requires magnetic probes to measure only the radial component of the magnetic field. This requirement is not satisfied in the TT-1 tokamak. Therefore, in this project, the Toroidal Filament Model has been adapted to account for the orientation of magnetic probes in circular tokamaks (Figure 3.1).

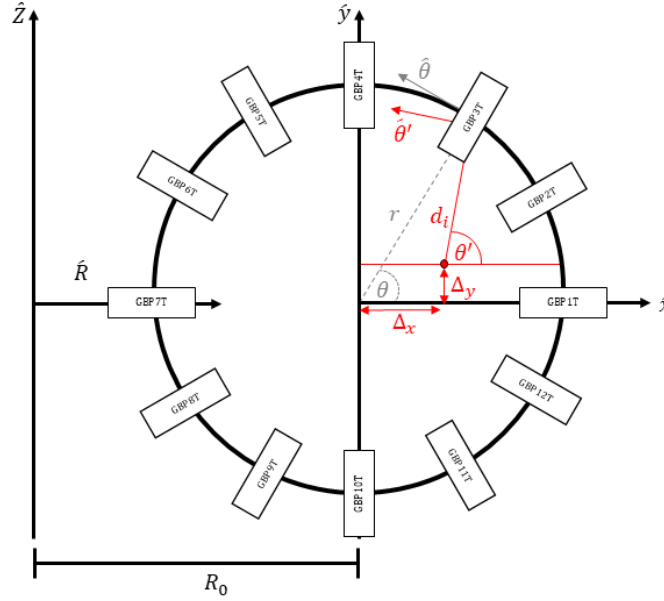


Figure 3.1: Poloidal plane of TT-1 showing magnetic probe configuration and plasma column position (Δ_R, Δ_Z)

In TT-1, the magnetic field generated by the plasma is measured by 12 magnetic probes—commonly referred to as Mirnov probes—uniformly installed around the vessel circumference. Photographs of the magnetic probes installed in the TT-1 tokamak, along with their installation locations, are shown in Figure 3.2. The probes are labeled

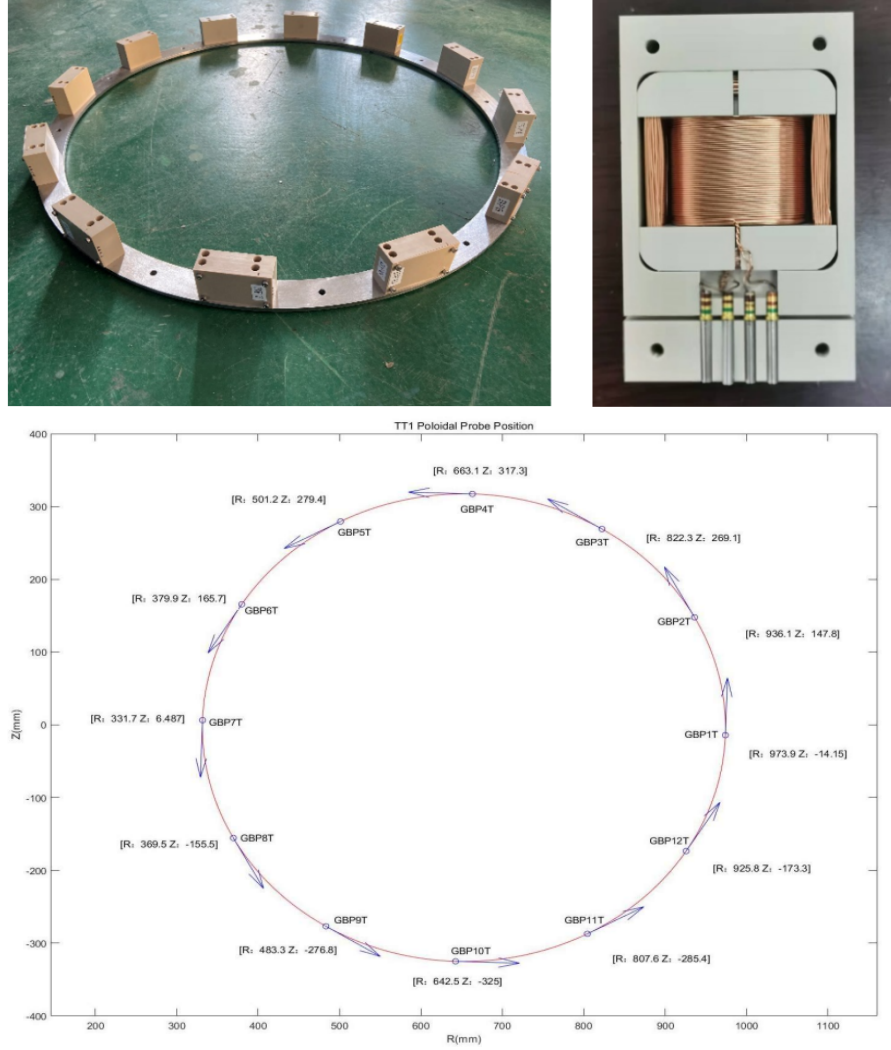


Figure 3.2: Photographs of a magnetic probe in the TT-1 tokamak and their installation locations

“GBP1T” through “GBP12T” and are positioned at an average radial distance $r = 0.321$ m from the center of the torus. They are evenly spaced at approximately 30° intervals, with GBP1T located at $\theta = 0^\circ$, as illustrated in Figure 3.1. Generally, the magnetic probes consist of induction coils wrapped around a core used to measure the scalar magnetic field component along their respective normal directions. Furthermore, the probes are also capable of detecting magnetohydrodynamic (MHD) fluctuations.

3.1.1 Straight-Line Current

The first step of the toroidal filament model involves analyzing a simplified scenario in which $R_0 \rightarrow \infty$, representing a first-order approximation of the actual configu-

ration. In this case, an infinite straight-line current I_0 , representing the plasma current, is assumed to pass through the poloidal plane at the point (Δ_x, Δ_y) , as illustrated in Figure 3.1.

According to Ampère's law, the magnetic field measured by each magnetic probe due to the current line located at (Δ_x, Δ_y) is given by

$$\mathbf{B}_i = \frac{\mu_0 I_0}{2\pi d_i} \hat{\theta}'. \quad (3.1)$$

where d_i denotes the distance from the straight-line current to magnetic probe i , and $\hat{\theta}'$ is the azimuthal unit vector defined with respect to the point (Δ_x, Δ_y) . Each magnetic probe is oriented such that it measures only the component of the magnetic field along its local $\hat{\theta}$ direction. Therefore, to obtain the magnetic field recorded by each probe (B_θ), the total magnetic field derived from Ampère's law must be projected onto the $\hat{\theta}$ axis. Using basic geometry, the unit vector $\hat{\theta}'$ can be expressed as

$$\hat{\theta}' = -\sin(\theta')\hat{i} + \cos(\theta')\hat{k} = \frac{1}{d_i} \left(-(y_i - \Delta_y)\hat{i} + (x_i - \Delta_x)\hat{k} \right). \quad (3.2)$$

The projection of the magnetic field onto the $\hat{\theta}$ axis is then

$$B_\theta = \mathbf{B}_i \cdot \hat{\theta}, \quad (3.3)$$

$$= \frac{\mu_0 I_0}{2\pi d_i^2} \left(-(y_i - \Delta_y)\hat{i} + (x_i - \Delta_x)\hat{k} \right) \cdot \left(-\sin(\theta)\hat{i} + \cos(\theta)\hat{k} \right), \quad (3.4)$$

$$= \frac{\mu_0 I_0}{2\pi d_i^2} \left((y_i - \Delta_y) \sin(\theta) + (x_i - \Delta_x) \cos(\theta) \right). \quad (3.5)$$

Given the setup of uniformly distributed magnetic probes around the torus circumference, the coordinates of each magnetic probe are given by

$$x_i = r \cos(\theta), \quad y_i = r \sin(\theta). \quad (3.6)$$

Thus, after algebraic manipulation, B_θ can be rewritten as

$$B_\theta = \frac{\mu_0 I_0}{2\pi} \times \frac{r - (\Delta_x \cos(\theta) + \Delta_y \sin(\theta))}{(r \sin(\theta) - \Delta_y)^2 + (r \cos(\theta) - \Delta_x)^2}. \quad (3.7)$$

In experimental application, a proportionality constant k_i must be multiplied to B_θ to obtain the magnetic signal measured by each probe (S_θ). Identical magnetic probes

share the same value of k_i [4]. Furthermore, since subsequent calculations involve ratios of magnetic signals, the term $k_i \frac{\mu_0 I_0}{2\pi}$ is normalized to simplify the equations:

$$S_\theta = \frac{r - (\Delta_x \cos(\theta) + \Delta_y \sin(\theta))}{(r \sin(\theta) - \Delta_y)^2 + (r \cos(\theta) - \Delta_x)^2}. \quad (3.8)$$

To analytically determine Δ_x and Δ_y , a small displacement assumption is made:

$$\frac{\sqrt{\Delta_x^2 + \Delta_y^2}}{R} \ll 1, \quad \frac{\Delta_x}{R \cos(\theta)} \ll 1, \quad \frac{\Delta_y}{R \sin(\theta)} \ll 1.$$

Keeping only first-order terms, the approximate form of the magnetic signal is given by (Appendix A):

$$S_\theta = \frac{1}{r} \times \left[1 + \frac{1}{r} (\Delta_x \cos(\theta) + \Delta_y \sin(\theta)) \right]. \quad (3.9)$$

To validate the earlier normalization, the signal recorded at the magnetic probe located at the opposite position, i.e., at $\theta + \pi$, is considered:

$$S_{\theta+\pi} = \frac{1}{r} \times \left[1 - \frac{1}{r} (\Delta_x \cos(\theta) + \Delta_y \sin(\theta)) \right]. \quad (3.10)$$

Equations 3.9 and 3.10 can be combined to form

$$S_P(\theta) \equiv \frac{S_\theta - S_{\theta+\pi}}{S_\theta + S_{\theta+\pi}} = \frac{1}{r} (\Delta_y \sin(\theta) + \Delta_x \cos(\theta)). \quad (3.11)$$

The resulting quantity S_P contains information about the plasma displacement. However, a single equation is insufficient to solve for displacements in both dimensions. Therefore, an additional pair of opposing magnetic probes located at θ_2 and $\theta_2 + \pi$ is used in conjunction with Equation 3.11 to uniquely determine the displacement along both axes. After algebraic manipulation, the displacements are given by:

$$\begin{bmatrix} \Delta_x \\ \Delta_y \end{bmatrix} = -\frac{r}{\sin(\theta_1 - \theta_2)} \times \begin{bmatrix} \sin(\theta_2) & -\sin(\theta_1) \\ -\cos(\theta_2) & \cos(\theta_1) \end{bmatrix} \begin{bmatrix} S_P(\theta_1) \\ S_P(\theta_2) \end{bmatrix}. \quad (3.12)$$

Thus, by using any two distinct pairs of opposing magnetic probes, the plasma displacement—under the assumption of an infinite straight-line current and small displacement—can be analytically solved. This concludes the adjustment for the circular magnetic probe configuration.

3.1.2 The Toroidal Filament B Field

Next, consider a toroidal filament forming a circular current loop in cylindrical coordinates carrying plasma current I_p with radius a_f (Figure 3.3). Assuming axisymmetry along the toroidal unit angle, the radius of the filament depends on the horizontal displacement of the plasma, such that $a_f = R_0 + \Delta_R$. Since the filament can also be vertically displaced by Δ_Z , the center of the toroidal loop does not necessarily coincide with the geometric center of the tokamak.

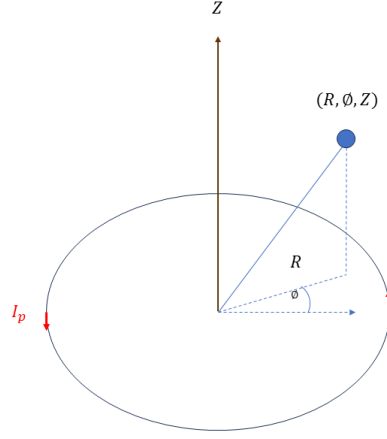


Figure 3.3: Toroidal filament with plasma current and magnetic field distribution at (R, ϕ, Z)

The motion of charges within the filament generates a magnetic field according to the Biot–Savart law. By placing the filament center at the origin and exploiting the polar symmetry, it suffices to evaluate the magnetic field at (R, ϕ, Z) :

$$\begin{aligned}
 B_r &= \frac{\mu_0 I_0}{2\pi} \frac{(Z + \Delta_Z)/R}{\sqrt{(a_f + R)^2 + (Z + \Delta_Z)^2}} \left(-K + \frac{a_f^2 + R^2 + (Z + \Delta_Z)^2}{(a_f - R)^2 + (Z + \Delta_Z)^2} E \right), \\
 B_z &= \frac{\mu_0 I_0}{2\pi} \frac{1}{\sqrt{(a_f + R)^2 + (Z + \Delta_Z)^2}} \left(K + \frac{a_f^2 - R^2 - (Z + \Delta_Z)^2}{(a_f - R)^2 + (Z + \Delta_Z)^2} E \right), \\
 B_\phi &= 0.
 \end{aligned} \tag{3.13}$$

Equation 3.13 is modified to incorporate the vertical displacement Δ_Z of the plasma, which accounts for every occurrence of the term $Z + \Delta_Z$ instead of the standard Z . Here,

K and E denote the complete elliptic integrals of the first and second kinds defined as

$$\begin{aligned} k &\equiv \sqrt{\frac{4a_f R}{(a_f + R)^2 + (Z + \Delta_Z)^2}}, \\ K(k) &= \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}, \\ E(k) &= \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta. \end{aligned} \quad (3.14)$$

3.1.3 Power Series Fitting

The straight-line current described in Subsection 3.1.1 serves as the first-order approximation of the actual plasma current, which forms a closed loop around the toroidal chamber. The magnetic signal derived from Equation 3.13 is used to connect this first-order approximation (Δ_x, Δ_y) to the actual plasma column position (Δ_R, Δ_Z) .

Since Δ_x and Δ_y represent first-order approximations of the true plasma column displacements, a power series expansion with m terms is employed to establish the relationship between them:

$$\begin{aligned} \Delta_R(t_n) &\approx \sum_{i=0}^m \beta_i(t_{n-1}) \Delta_x(t_n), \\ \Delta_Z(t_n) &\approx \sum_{i=0}^m \alpha_i(t_{n-1}) \Delta_y(t_n). \end{aligned} \quad (3.15)$$

For the current implementation, m is set to 3. Furthermore, the uncertainty in Δ_R and Δ_Z can be calculated using the error propagation formula where $\mathbf{C}$ is the covariance matrix of the coefficients:

$$\sigma_q = \sqrt{\mathbf{v}^T \mathbf{C} \mathbf{v}}; \quad \mathbf{v} = [q^0, \dots, q^m]. \quad (3.16)$$

Using Equation 3.15, the plasma column position at time step t_n can be computed from the first-order approximation and the power series coefficients α_i and β_i . Note that the coefficients are determined based on the preceding time step t_{n-1} , as in real-time experimental scenarios it is not feasible to directly fit the values of Δ_x and Δ_y to the unknown Δ_R and Δ_Z . The assumption of a small time interval between t_{n-1} and t_n justifies the approximation of minimal displacement over that interval, allowing the use

of nearly constant coefficients. Consequently, the displacement at the initial time t_0 , when the plasma becomes ionized (discharge time), is set to $(\Delta_R, \Delta_Z) = (0, 0)$.

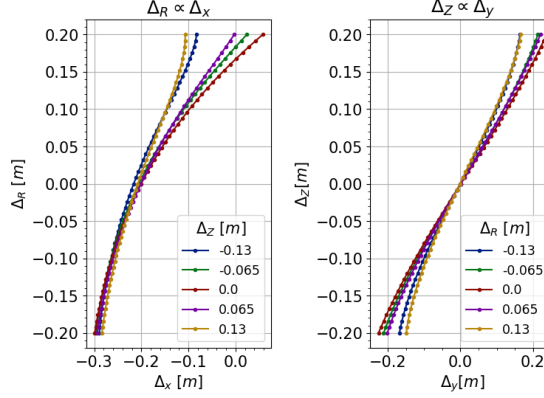
The coefficients α_i and β_i are obtained by evaluating the magnetic field at each probe using Equation 3.13 and then using those results to compute Δ_x and Δ_y via Equation 3.12. Although Equation 3.13 appears complex, it fundamentally depends on only two independent variables: Δ_R and Δ_Z . With this dependency, the magnetic field at any probe can be calculated and substituted into Equation 3.12, yielding the mapping from the first-order approximation (Δ_x, Δ_y) to the actual plasma column position (Δ_R, Δ_Z) :

$$\Delta_x = f(\Delta_R, \Delta_Z), \quad \Delta_y = f(\Delta_R, \Delta_Z).$$

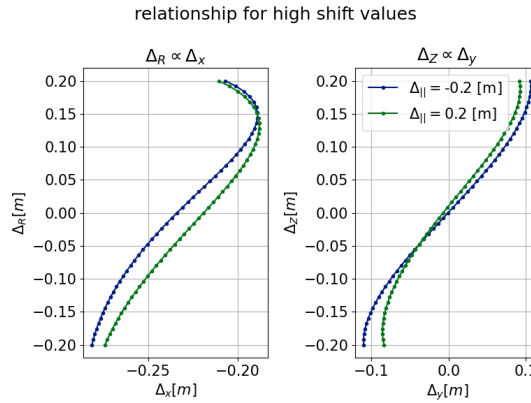
To enable fast and reliable calculations during experiments, the coefficients of the power series are precomputed using simulated values of Δ_R and Δ_Z . The computation of the coefficients follows these steps:

1. For given Δ_R and Δ_Z , calculate the magnetic field signal at selected magnetic probes using Equation 3.13.
2. Using the magnetic field signal from Step 1, calculate the corresponding Δ_x and Δ_y with Equation 3.12.
3. Repeat Steps 1 and 2 for other values of Δ_R and Δ_Z to obtain the relationships $\Delta_R \propto \Delta_x$ (given Δ_Z) and $\Delta_Z \propto \Delta_y$ (given Δ_R).
4. Fit power series of order m to both relationships to obtain the coefficients α_i and β_i .

Using the set of magnetic probes GBP1T, GBP4T, GBP7T, and GBP10T shown in Figure 3.2, the relationship shown in Figure 3.4 was obtained. In this project, Equation 3.12 allows the use of any two sets of opposing magnetic probes. Since there exist six opposing probe pairs in the TT-1 tokamak, there are $\binom{6}{2} = 15$ possible combinations. Each set corresponds to its own unique set of coefficients.

Figure 3.4: Relationships between $\Delta_R \sim \Delta_x$ and $\Delta_Z \sim \Delta_y$

One key point to consider is that the relationships $\Delta_R \sim \Delta_x$ and $\Delta_Z \sim \Delta_y$ are valid only within a limited range. Since Δ_R and Δ_Z are numerically computed as the inverse of the first-order approximation, a single value on the x -axis can correspond to multiple values on the y -axis (see Figure 3.5). At large displacements, data points fold back, degrading curve-fitting accuracy. This folding begins at the local maximum,¹ where $\frac{d\Delta_x}{d\Delta_R} = 0$. The probe set with the smallest distance to this local maximum from $\Delta_R = 0$ defines the valid domain for applying the toroidal filament model, estimated to be within ± 0.1 , m.

Figure 3.5: Example relationship of $\Delta_R \sim \Delta_x$ and $\Delta_Z \sim \Delta_y$ for high displacement values

¹In experiments, Δ_Z tends to be small and is therefore not used to calculate the shifting domain.

3.1.4 Toroidal Filament Model in Experimental application

Once the coefficients are computed for each magnetic probe set, the plasma column position can be determined in real-time applications. For efficient implementation, the coefficients are pre-stored in a hash map, eliminating the need for expensive curve fitting, enabling constant-time complexity. The calculation process is as follows. At the beginning of the experiment, magnetic probes measure the magnetic field generated by the plasma. This information is used to calculate the first-order displacements Δ_x and Δ_y via Equation 3.12. The values of Δ_R and Δ_Z from the previous time step t_{n-1} are then used to retrieve the relevant coefficients for the selected probe set. These coefficients, along with the first-order displacements, are applied to Equation 3.15 to compute the plasma position at t_n (Figure 3.6).

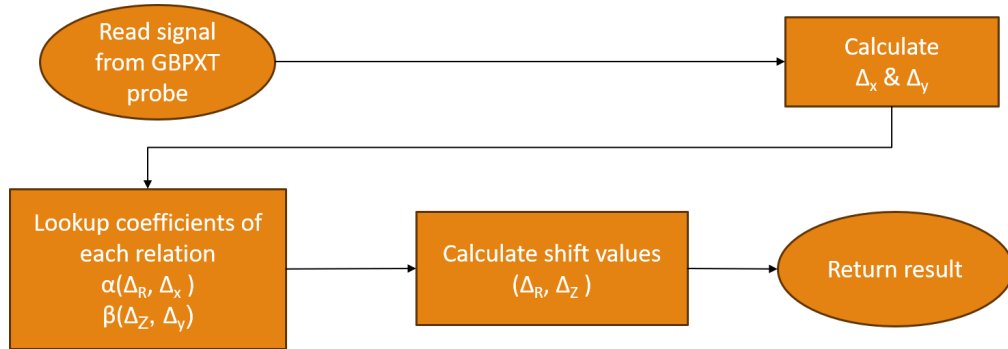


Figure 3.6: Flowchart of toroidal filament model implementation in real-time experiments

3.2 Optical Boundary Reconstruction (OFIT)

Figure 1.3 illustrates the TT-1 tokamak structure and plasma, highlighting key challenges in applying OFIT. These challenges arise primarily from empty equipment ports (black circular surfaces in the image) and the relatively low plasma temperature compared to larger tokamaks. The lower plasma temperature increases visible spectrum emission, resulting in stronger light reflections from structural surfaces and introducing significant noise. Additionally, the presence of numerous empty ports complicates edge detection. Consequently, effective noise reduction is essential for successful application of OFIT in TT-1.

Figure 1.1 shows a schematic of the TT-1 tokamak. In the center panel, the CCD camera is visible in the lower right corner. This camera is oriented toward the tokamak's interior to capture plasma images. At the time of this project, methods for accurately measuring the CCD camera's extrinsic parameters in TT-1 are still under development. Therefore, approximate parameters are used in this project. The parameters adopted are listed in Figure 3.7.

Extrinsic Parameters	
Camera location (u_c, v_c, w_c) [m]	$(0, -0.036, 0.985)$
Angle between optical axis and sagittal plane (θ_s)	32°
Angle between optical axis and transverse plane (θ_t)	0°
Intrinsic Parameters	
Focal length f [mm]	12
Scaling factor K_{scale}	2.5×10^{-4}
Distortion (TV)	5.9×10^{-3}

Figure 3.7: Parameters of CCD camera used in this project

3.2.1 Defining ROIs

The first major step in OFIT is plasma edge detection. As previously mentioned, the plasma edge is assumed to correspond to pixels with the highest relative brightness. While this process is generally straightforward, processing every pixel in a 1920×1080 image is computationally expensive. To reduce computation time and isolate the edge on each side of the plasma, two circular regions of interest (ROIs) are defined—one on the

high-field side and one on the low-field side². The circular shape preserves the inherent circularity of the plasma edge (Figure 3.8).

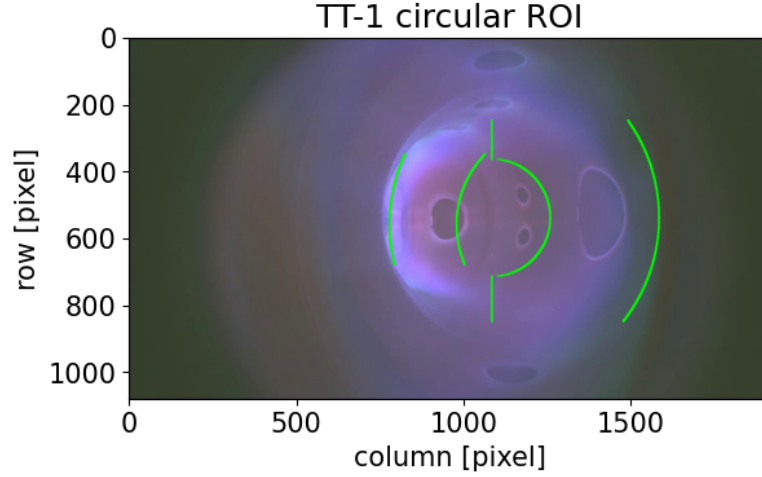


Figure 3.8: Circular ROIs defining the search domain

3.2.2 Image Processing

The next step involves processing the image in preparation for edge detection. The following steps are applied:

1. Image undistortion to remove lens distortion.
2. Gaussian blur to reduce random noise.
3. Conversion to grayscale.
4. Application of edge enhancement kernel.
5. (Optional) image masking.

Most steps are standard, except for edge enhancement and masking. In raw plasma images, the plasma edge is often difficult to detect. A convolution kernel (Figure 3.9) is applied to increase intensity contrast along the edge. For each pixel, a weighted sum of neighboring pixels is used to enhance the central pixel's intensity, thereby improving edge visibility.

²The high-field side refers to the inner side of the torus, while the low-field side refers to the outer side.

1	2	3	0	-3	-2	-1
1	2	3	0	-3	-2	-1
1	2	3	0	-3	-2	-1

Figure 3.9: Convolution filter kernel used for edge enhancement

Image masking is optional but useful when the plasma edge appears over empty ports, where pixel intensity may decrease due to line integration. This can cause false detections. To mitigate this, the HSV color range of the plasma is identified and masked from the grayscale image. This technique is manual and plasma-dependent making it an optional step. The defined HSV range for TT-1 plasma is:

$$40 < H < 140, \quad S > 200, \quad V > 30$$

3.2.3 Port Detection

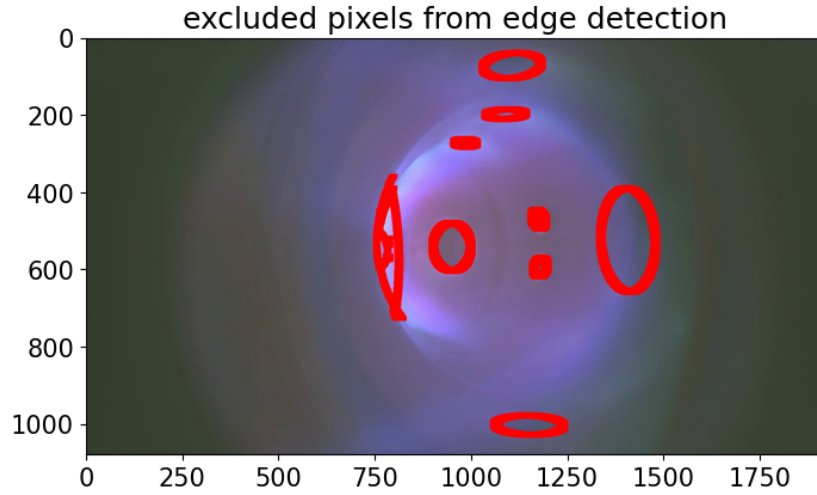


Figure 3.10: Detected port pixels excluded from plasma edge detection

Empty ports not only reduce plasma intensity but also introduce strong, spurious edges due to reflection (Figure 1.3). These appear as bright arcs, interfering with detection. To prevent this, port edges are identified and excluded using an ellipse-fitting algorithm [10]. Figure 3.10 shows the detected port-related pixels excluded from edge detection.

3.2.4 Plasma Edge Detection Algorithm

With ROIs defined, the image preprocessed, and port structures identified, the plasma edge detection algorithm proceeds. A localized window search is used to limit computation and increase robustness. The flowchart is shown in Figure 3.11.

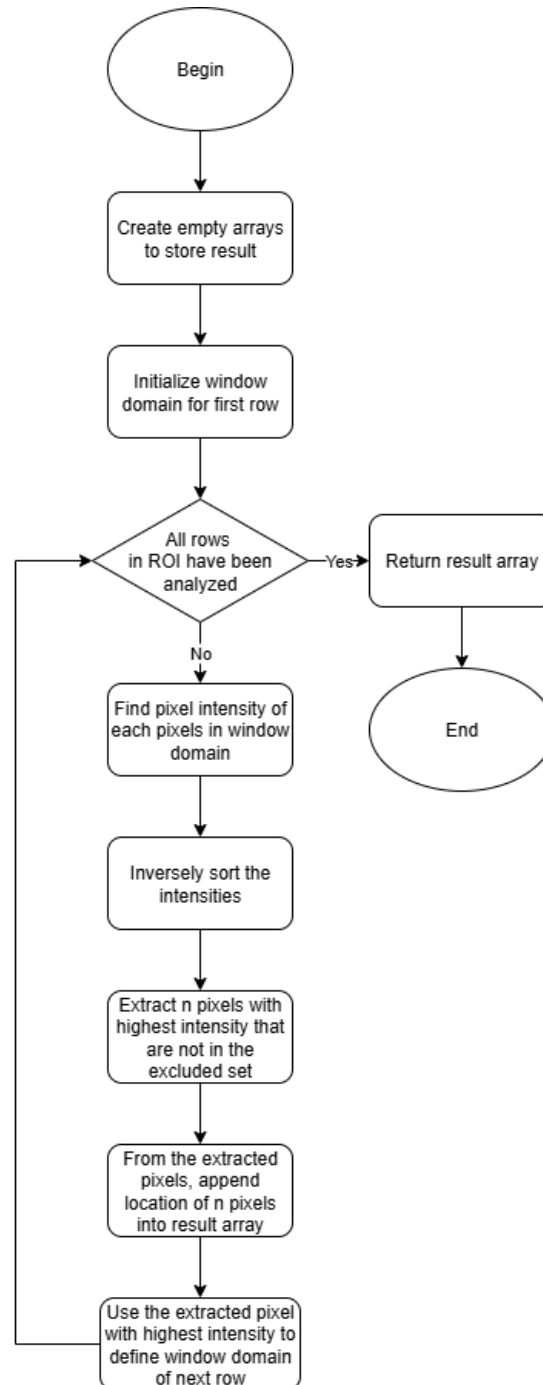


Figure 3.11: Flowchart of the plasma edge detection algorithm (python implementation in `find_edge` function Github `OFIT_dir/projection_detection.py`)

The algorithm begins by creating an empty array to store detected edge pixels (x_e, y_e) . In the first row, the entire ROI is searched. For each subsequent row, the search is limited to a window centered around the highest intensity pixel from the previous row. In each row, grayscale intensities are sorted and the top n brightest pixels that do not overlap with masked or port regions are selected. The brightest pixel $(x_{\max, y})$ defines the window for the next row $(y + 1)$ with a window of size d :

$$\text{Window domain of row } y+1 = [x_{\max, y} - d, x_{\max, y} + d]. \quad (3.17)$$

Figure 3.12 shows the result of this edge detection process.

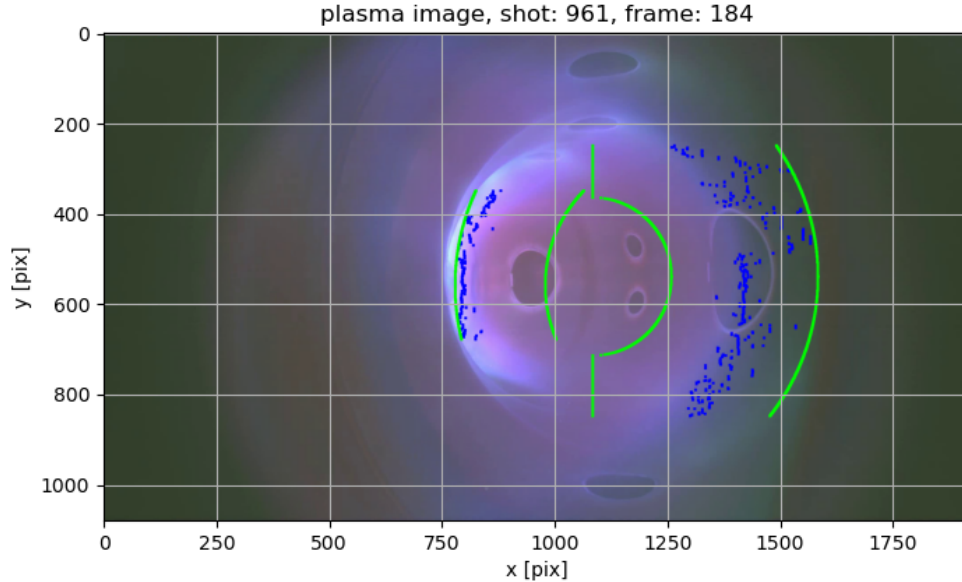


Figure 3.12: Detected plasma edge pixels (highlighted in blue)

3.2.5 Projection Plane Transformation

Because pixel coordinates are two-dimensional, their transformation is restricted to a 2D surface in 3D space. Thus, all detected edge pixels are assumed to lie on a projection plane $(u, v, 0)$.

To transform (x_e, y_e) into projection plane coordinates $(u_e, v_e, 0)$, the camera's

projection matrix P is used:

$$\begin{bmatrix} \tilde{x}_e \\ \tilde{y}_e \\ \tilde{z} \end{bmatrix} = P \begin{bmatrix} u_e \\ v_e \\ 0 \\ 1 \end{bmatrix}. \quad (3.18)$$

Here, $(\tilde{x}_e, \tilde{y}_e, \tilde{z})$ are the homogeneous pixel coordinates, and P encapsulates the camera's intrinsic and extrinsic parameters (Appendix B).

3.2.6 Poloidal Plane Transformation

Under the assumption of toroidal symmetry, the plasma edge points projected onto the $(u, v, 0)$ plane can be further transformed into poloidal coordinates (R_e, Z_e) by calculating the azimuthal angle ϕ_e of each point. The mathematical formulation used for this transformation is discussed in Section 2.2, specifically Equation 2.3.

This transformation requires evaluating the curvature of the plasma edge in the projection plane via the derivative $\frac{du_e}{dv_e}$. To estimate this derivative, the edge points within each ROI are fitted with a horizontal parabola:

$$u_e = av_e^2 + bv_e + c. \quad (3.19)$$

Although ellipse fitting was also considered, it introduced computational overhead and ambiguity in orientation (see Figure 3.13(b)). The parabola fitting is simpler and, when combined with a robust outlier detection algorithm, sufficiently accurate for this purpose.

To mitigate the effect of noise on curve fitting, the RANSAC algorithm (Appendix C) is used. It is preferred over the Hough transform due to its faster execution and fewer required parameters. Fitting result from RANSAC is shown in Figure 3.13(a).

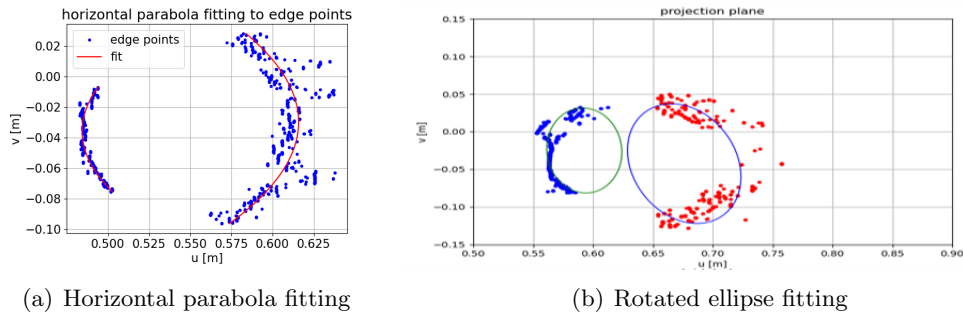


Figure 3.13: Comparison of horizontal parabola and rotated ellipse fitting

Once the curvature is estimated, Equation 2.3 is applied to transform each edge point into poloidal coordinates. A circle is then fitted to the (R_e, Z_e) points. The center (R_c, Z_c) of this circle defines the plasma column displacement $(\Delta_R, \Delta_Z) = (R_c - R_0, Z_c)$, and the radius corresponds to the plasma's minor radius. Figure 3.14 shows the result of the OFIT transformation based on the detected edge pixels shown in Figure 3.12, for experimental shot 961 at frame 184.

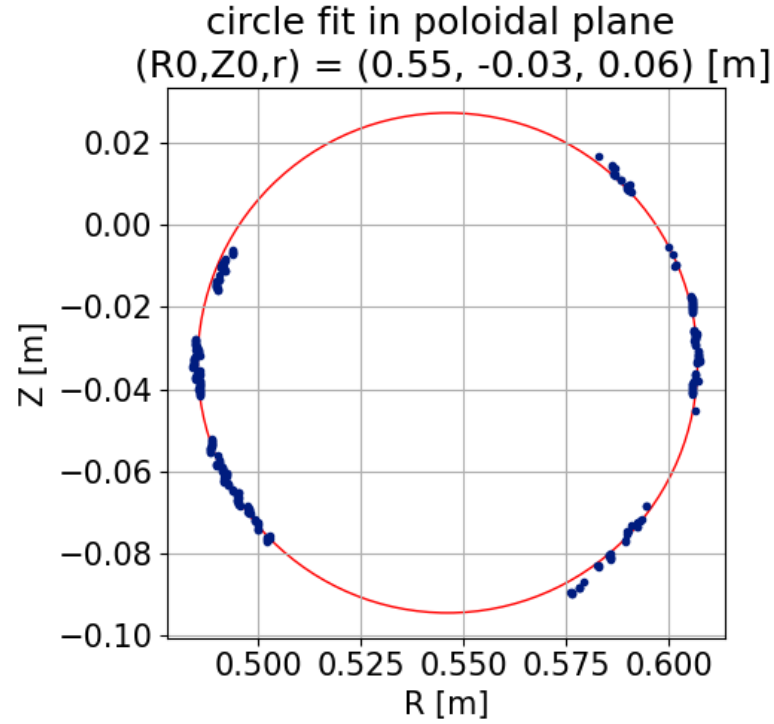


Figure 3.14: Plasma shape from OFIT for shot 961 at frame 184 on the poloidal plane, with RANSAC fitting.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Uncertainty and Computation Time from Simulation

To evaluate the uncertainty and computation time of the toroidal filament model and Optical Boundary Reconstruction (OFIT), simulations were conducted using a Python implementation running on a machine equipped with an Intel Core i5 processor at 1.60 GHz and 8 GB of RAM. A time series of simulated toroidal filaments was generated by introducing random perturbations to the displacement at each time step, starting with zero displacement at the initial step. These simulated displacements were used to generate synthetic magnetic signals at each magnetic probe. Applying the toroidal filament model to this simulated data, the mean errors across all probes for radial displacement (Δ_R) and vertical displacement (Δ_Z) were found to be 0.6mm and 0.3mm, respectively, within a displacement range of 10 cm. The average computation time per probe set was approximately 97.1 μ s. These results demonstrate the model's fast computation and low error, supporting its potential for real-time application.

The error and runtime of OFIT were assessed using a simulated plasma shell with a circular cross-section on the projection plane. The simulated plasma edge points (u_e, v_e) and azimuthal angle ϕ_e were generated using the equations derived by Hommen et al. [7]:

$$\begin{aligned}
 R_e &= R_0 + r \cos(\theta), \quad \theta \in [0, 2\pi], \\
 Z_e &= r \sin(\theta), \\
 \sqrt{u_c^2 + w_c^2} \sin\left(\phi_e + \arctan\left(\frac{u_c}{w_c}\right)\right) &= R_e - (Z_e - v_c) \frac{dR_e}{dZ_e}, \\
 u_e - u_c &= w_c \frac{R_e \cos(\phi_e) - u_c}{w_c - R_e \sin(\phi_e)}, \\
 v_e - v_c &= w_c \frac{Z_e - v_c}{w_c - R_e \sin(\phi_e)}.
 \end{aligned} \tag{4.1}$$

Applying OFIT to this simulated plasma shell, the mean errors along Δ_R and Δ_Z were $90\ \mu\text{m}$ and $80\ \mu\text{m}$, respectively. These minimal errors indicate that the transformation process introduces negligible inaccuracies, with the primary source of error likely arising from the edge detection procedure. However, quantifying edge detection error is challenging due to its dependence on input image noise. The average runtime of the OFIT algorithm, using 30 RANSAC iterations per frame, was 0.37 ± 0.07 seconds.

4.2 Application to Experimental Data

With estimated errors established, both the toroidal filament model and OFIT were applied to experimental data from the TT-1 tokamak. Accurate application of the toroidal filament model requires effective elimination of noise magnetic fields generated by the device and eddy currents induced within the reactor vessel. In contrast, OFIT primarily depends on accurate knowledge of the camera's projection matrix, obtained via camera calibration (see Appendix B). As of June 2025, however, the projection matrix for the TT-1 CCD camera has not been precisely determined; thus, OFIT results presented herein are based on approximate projection parameters.

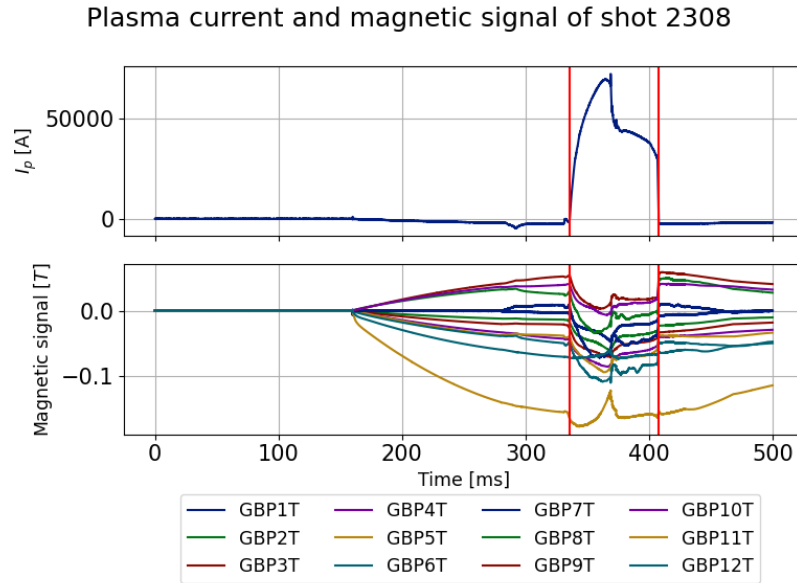


Figure 4.1: Plasma current and magnetic signals from each probe during TT-1 experimental shot 2308

During experiments, a Rogowski coil measures the plasma current (I_p), while magnetic probes record magnetic signals at the positions shown in Figure 3.2. Figure 4.1 displays the raw plasma current and magnetic signals for all probes. Vertical red lines indicate the start and end of plasma discharge. The discharge duration is defined as the interval during which the plasma current remains above 2500 A; the start of this interval is referred to as the discharge time.

Since the toroidal filament model assumes that the magnetic field arises solely from the toroidal filament, magnetic contributions from other sources must be removed [5]. In this work, magnetic fields measured at the discharge start time are set to zero to correct for coil contributions. Figure 4.2 shows the corrected magnetic signals during the discharge after applying this method. However, because the tokamak vessel is constructed from conducting materials, eddy currents induced in the vessel, which are proportional to the plasma current, present a more challenging source of noise. Figure 4.3 illustrates the expected offset in displacement (Δ_x) caused by varying peak plasma current. Multiple approaches, both hardware- and software-based, have been proposed to mitigate vessel eddy effects [6, 11–14].

Plasma current and corrected magnetic signal of shot 2308

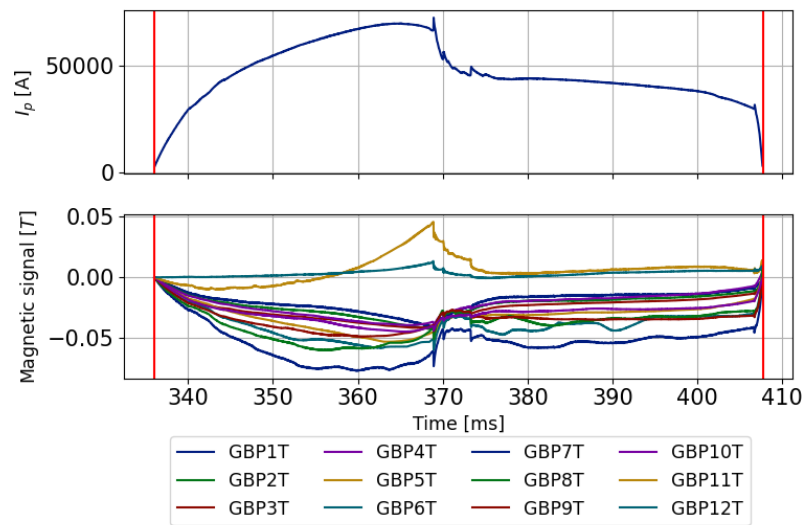


Figure 4.2: Magnetic signals corrected using measurements at discharge start time

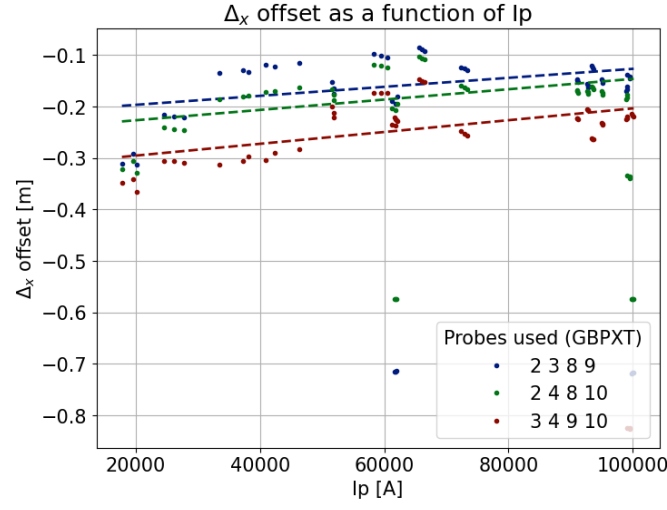


Figure 4.3: Magnetic field offset as a function of plasma current

After noise correction, the toroidal filament model was applied for plasma column position calculations. However, TT-1 currently lacks a robust method to eliminate vessel eddy effects, so results may be influenced by induced currents. The initial magnetic signals at discharge time were used as baselines to remove coil contributions. Figure 4.4 presents the plasma column positions calculated using magnetic probes GBP1T, GBP4T, GBP7T, and GBP10T. At discharge start, the plasma is radially displaced by approxi-

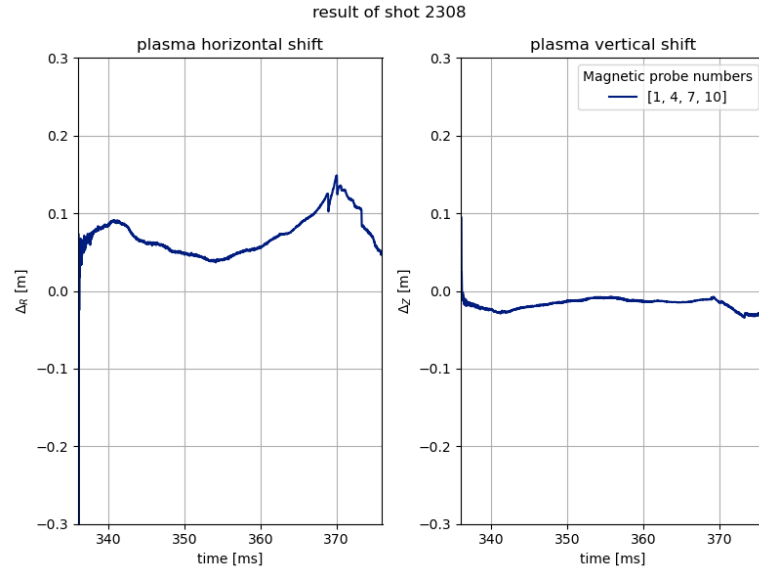


Figure 4.4: Plasma column position calculated from probes GBP1T, GBP4T, GBP7T, and GBP10T

mately 0.05 m. It then moves radially outward to nearly 0.1 m at 342 ms, before shifting inward until 355 ms and finally moving outward to contact the vessel on the low-field side at approximately 370 ms. This vessel contact is characterized by a displacement peak and subsequent sharp decrease. These times correlate with CCD camera images and a significant drop in plasma current (Figure 4.2). Vertically, the plasma stabilizes near zero displacement until vessel contact, when a sudden negative displacement occurs. The vertical displacement is less volatile than the radial, consistent which is a known behavior of plasma within TT-1.

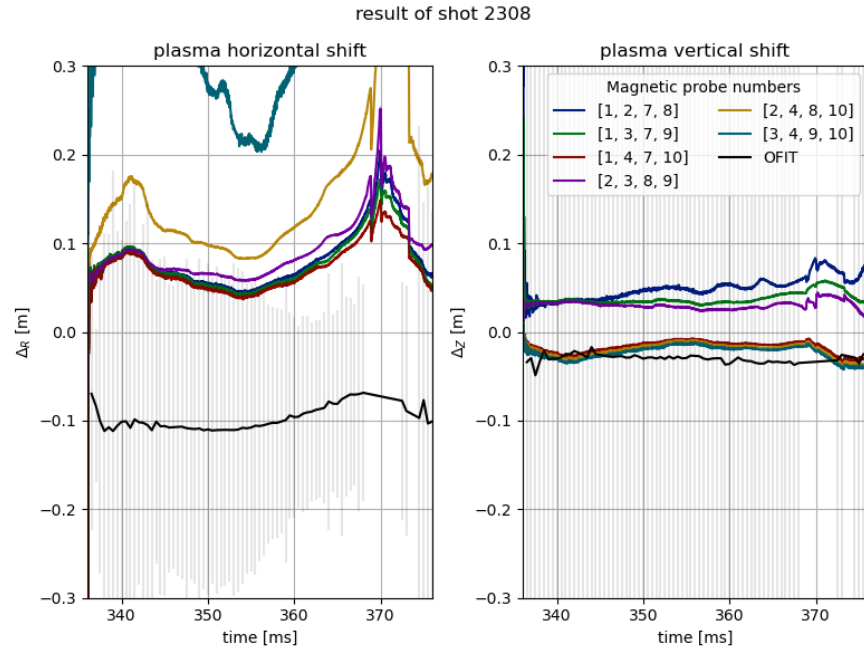


Figure 4.5: Plasma column position calculated via the toroidal filament model and OFIT

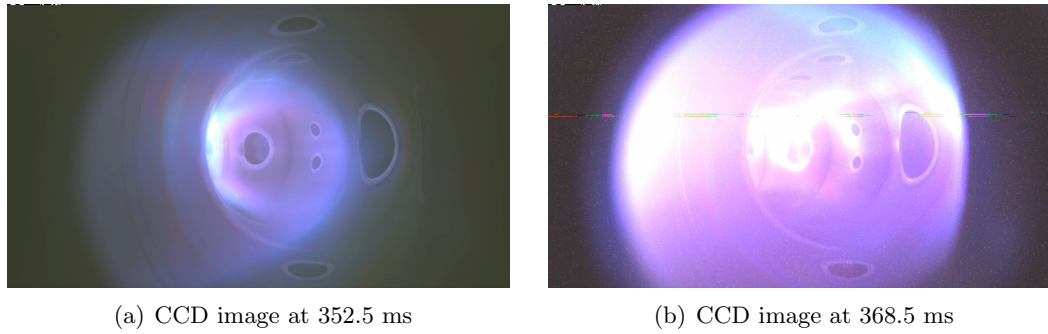


Figure 4.6: CCD images of shot 2308 illustrating plasma position at key times

In Figure 4.5, each color corresponds to displacement results from different probe combinations, along with OFIT results. Vertical gray lines indicate OFIT uncertainties. Errors from the toroidal filament model are plotted but not visible due to their small covariance. Probe sets [GBP3T, GBP4T, GBP9T, GBP10T] and [GBP2T, GBP4T, GBP8T, GBP10T] show radial displacement deviations likely caused by vessel eddy magnetic induction. The set [GBP1T, GBP4T, GBP7T, GBP10T] aligns better radially but shows discrepancies in Δ_Z , also probably due to eddy effects. OFIT results deviate significantly in radial displacement due to uncalibrated projection parameters.

Although the results obtained from the toroidal filament model and OFIT do not quantitatively agree, they both demonstrate similar displacement trends, such as comparable rates of change in displacement over time. Notably, at approximately 370 ms, the uncertainties of OFIT are not shown because, during this period, the plasma is either extremely close to or in contact with the vessel, positioning it too near the camera for edge detection by OFIT. Figure 4.6(a) presents a CCD image from experimental shot 2308 captured at 352.5 ms, a moment when the plasma is visually observed to have its maximum negative radial displacement. In contrast, Figure 4.6(b) shows the plasma at 368.5 ms, just before it impacts the vessel. These two images correspond closely with the toroidal filament model results, which calculate the plasma's maximum negative radial displacement at approximately 355 ms and maximum positive radial displacement at 370 ms. This strong agreement between the toroidal filament model, OFIT, and CCD images highlights the potential for these two methods to be effectively used for cross-validation, assuming proper removal of noise from various sources and accurate determination of projection parameters.

4.3 Limitation and Causes of Error

Although both methods show accurate results on simulated data, practical limitations affect their experimental application. The toroidal filament model is constrained by the range of plasma displacement it can handle and the effectiveness of magnetic noise

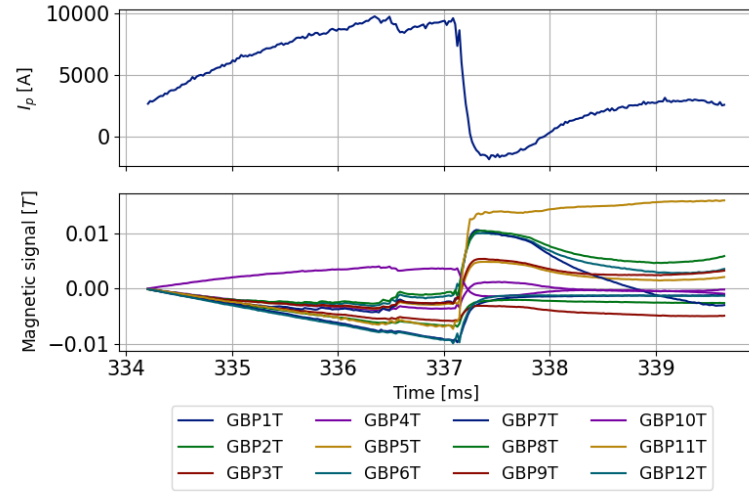
reduction. OFIT is limited by slow computation, camera vibrations, and the tokamak's reflective internal structures, which cause false plasma edge detections.

Magnetic noise and restricted displacement domains are the primary error sources for the toroidal filament model. As shown in Figure 4.3, induced magnetic fields from vessel eddy currents increase with plasma current and affect results. While the power series coefficients used in the model do not depend directly on plasma current, accurate removal of eddy-induced fields is essential. The imposed displacement domain (Section 3.1.3) limits radial displacement range; simulations show errors increase significantly when displacement exceeds this domain.

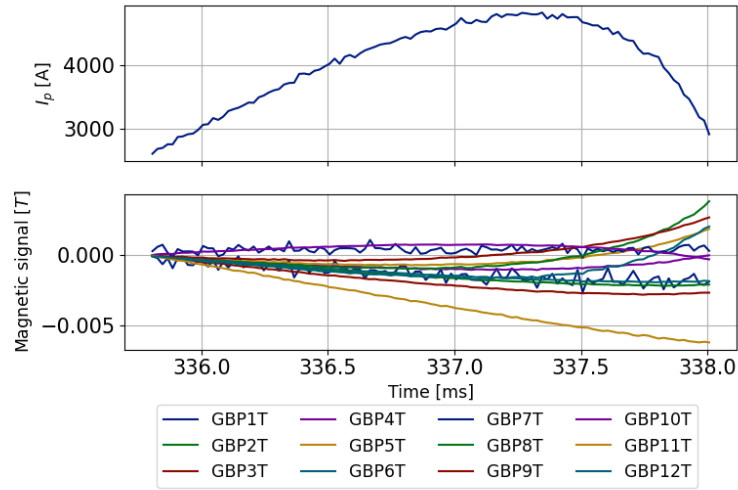
Figures 4.7 and 4.8 present plasma current, magnetic signals, and respective model results for various shots. Shots 370 and 928 exhibit low plasma currents, leading to high noise-to-signal ratios that degrade the toroidal filament model's performance. In shot 928, some magnetic signals do not correlate with plasma current, indicating noise contamination. Low plasma current also limits OFIT, as weak optical emission reduces plasma edge visibility, as seen in shots 370 and 928. From 19 experimental shots analyzed, both models are reliably applicable only when plasma current exceeds 50 kA.

The main causes of error for optical boundary reconstruction in the TT-1 tokamak are the reflective internal structure and camera movement. Because the plasma in TT-1 operates at relatively low temperatures, it emits substantial visible light. In some frames, this emission reflects off internal tokamak structures, producing false detections—particularly on the high-field side when the plasma displacement is large toward that region. Regarding camera movement, vibrations from tokamak operation can cause shifts in the CCD camera position and orientation, leading to uncertainty in the camera parameters needed for coordinate transformation. To address this, real-time camera calibration methods can be employed to continuously track the camera position throughout experiments. An implementation of such an algorithm is described in [8], where alternative edge detection techniques using line-segment detection and parallel image processing are also presented to reduce OFIT computation time.

Plasma current and corrected magnetic signal of shot 370



Plasma current and corrected magnetic signal of shot 928



Plasma current and corrected magnetic signal of shot 966

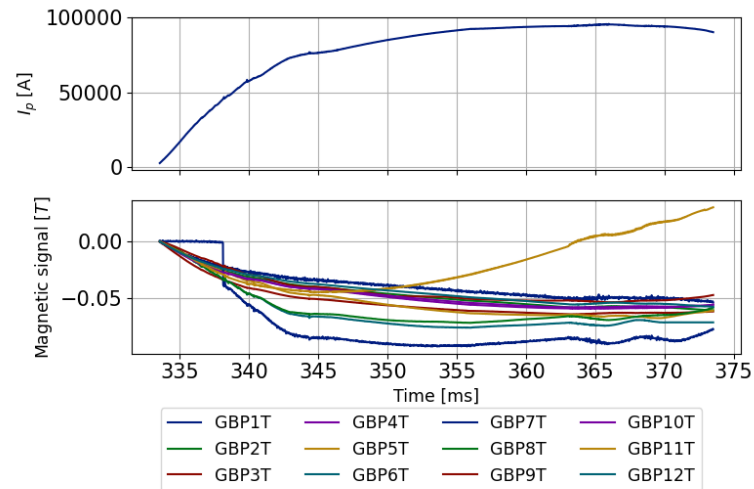


Figure 4.7: Plasma current and magnetic signal of different experimental shots

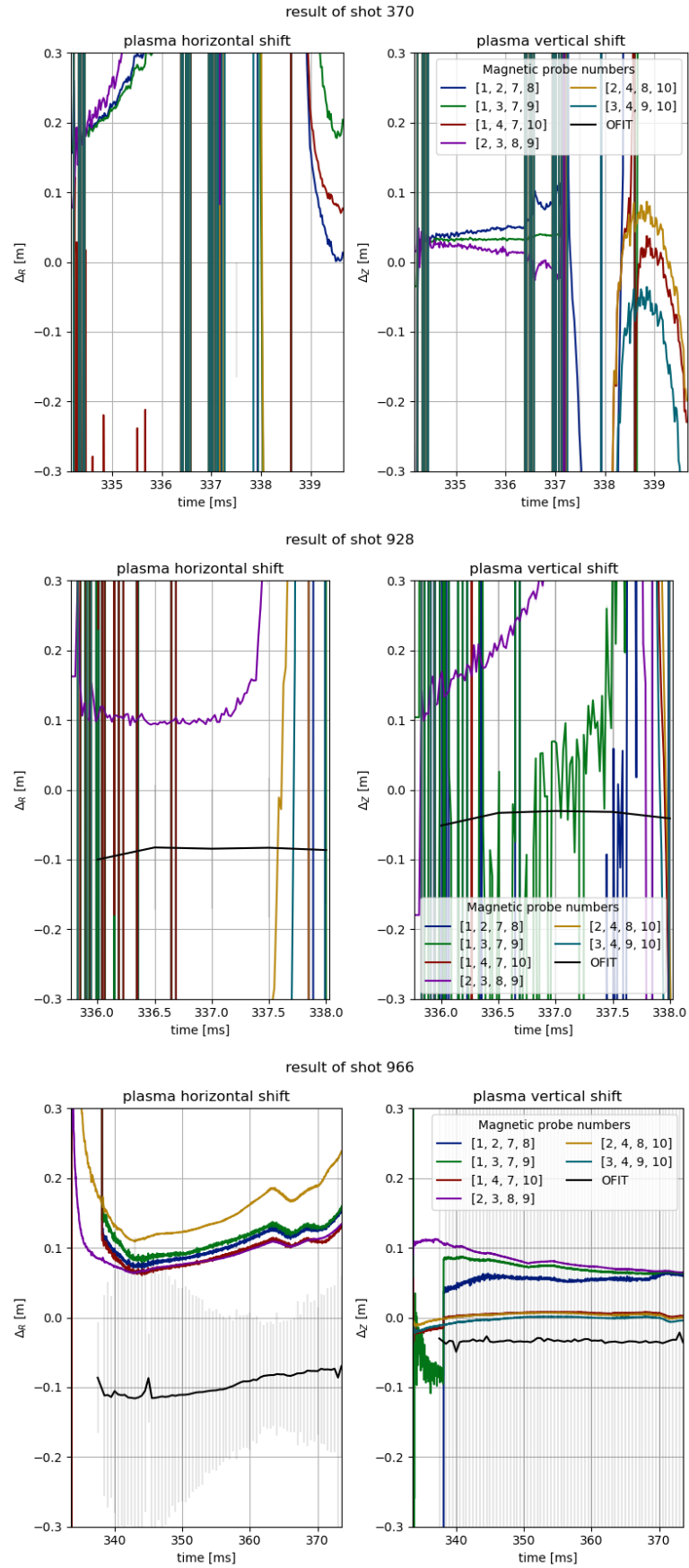


Figure 4.8: Result of toroidal filament model and OFIT on different experimental shots

CHAPTER 5

CONCLUSION AND FUTURE WORK

To stably confine plasma within a tokamak, accurate knowledge of the plasma column position is essential for providing real-time feedback control of the magnetic configuration. This thesis has presented the Toroidal Filament Model and Optical Boundary Reconstruction (OFIT) as two methods for determining the plasma column position in the TT-1 tokamak. The Toroidal Filament Model was adapted for use with the circular magnetic probe configuration of TT-1. It assumes a δ -function plasma density profile and reconstructs plasma position from magnetic probe signals. In contrast, OFIT estimates the plasma edge by detecting row-wise intensity maxima in CCD images, which are then transformed to the poloidal plane using camera projection parameters. To address the significant visual noise in TT-1 images—arising from reflections, ports, and low-temperature plasma—OFIT incorporates pixel exclusion, localized window searches, and RANSAC fitting to improve edge detection robustness. Together, these adaptations make both methods compatible with TT-1’s unique diagnostic limitations and operational environment.

Simulation results indicate that both methods yield minimal errors under idealized conditions. For displacements within ± 10 cm, the Toroidal Filament Model achieves mean position errors of 0.6mm in Δ_R and 0.3mm in Δ_Z , with a computation time of approximately 97.1 μ s per time step, supporting its suitability for real-time application. For OFIT, when applied to synthetic plasma shells, the coordinate transformation process—excluding the undetermined contribution of edge detection errors—achieves mean positional accuracy of 90 μ m in Δ_R and 80 μ m in Δ_Z , with a runtime of approximately 0.37 s per frame using 30 RANSAC iterations. The low uncertainty from OFIT transformation suggests that the primary source of error in OFIT arises not from the

transformation itself, but from the accuracy of edge detection, which is sensitive to image noise, reflections, and the clarity of the plasma boundary. These results validate the accuracy and performance of both methods in a controlled setting.

Although plasma current does not theoretically limit the applicability of the Toroidal Filament Model and OFIT, experimental limitations arise from the noise-to-signal ratio and plasma edge visibility. Analysis of 19 experimental shots—with plasma current ranging approximately from 4 kA to 100 kA—suggests that a current threshold of about 50 kA is necessary to maintain a low noise-to-signal ratio and observable plasma edge, allowing both methods to operate effectively.

Experimentally, the Toroidal Filament Model was applied using magnetic signals from TT-1 campaigns to estimate plasma column positions. A major source of error is magnetic noise induced by eddy currents within the tokamak vessel. Therefore, the model's accuracy depends heavily on effective mitigation of this magnetic noise.

From plasma images, potential error sources of OFIT were identified, particularly false plasma edge detection due to noise pixels. To reduce noise effects, a windowing approach and the RANSAC algorithm were employed for localized edge detection. However, under conditions of high noise—especially reflections from internal structures on the high-field side—OFIT was not fully sufficient.

Comparison of OFIT results with those from the Toroidal Filament Model reveals agreement in the overall trend of plasma displacement, though exact quantitative agreement is currently hindered by magnetic noise and the lack of precise camera projection parameters. Nonetheless, the similarity in trends indicates that both methods can be used reliably for plasma position determination in experimental settings, provided that noise is adequately suppressed and camera calibration is accurate.

Outlook to future improvements include accelerating OFIT computation time through alternative edge detection techniques (such as line-segment detection) along with parallel image processing. For the Toroidal Filament Model, extensions to include other probe configurations beyond opposing sets can be explored. Additionally, more

comprehensive magnetic reconstruction methods, such as those based on solutions to the Grad–Shafranov equation, offer promising avenues for capturing more detailed plasma characteristics.

APPENDIX

APPENDIX A

APPROXIMATION OF MAGNETIC SIGNAL

Starting from Equation 3.8, factor out the multiplication by R from the denominator:

$$\begin{aligned} S_\theta &= \frac{r - (\Delta_x \cos(\theta) + \Delta_y \sin(\theta))}{(r \sin(\theta) - \Delta_y)^2 + (r \cos(\theta) - \Delta_x)^2}, \\ &= \frac{1}{r^2} \times \frac{r - (\Delta_x \cos(\theta) + \Delta_y \sin(\theta))}{(\sin(\theta) - \frac{\Delta_y}{r})^2 + (\cos(\theta) - \frac{\Delta_x}{r})^2}. \end{aligned}$$

Assuming small displacements along each axis, terms of order higher than first in Δ_x and Δ_y are neglected, i.e., $(\Delta_x/r)^2 \approx 0$ and $(\Delta_y/r)^2 \approx 0$. The approximation of small displacement:

$$\frac{\Delta_x}{r \cos(\theta)} \ll 1, \quad \frac{\Delta_y}{r \sin(\theta)} \ll 1,$$

allows the signal equation to be written as

$$S_\theta = \frac{1}{r^2} \times \frac{r - (\Delta_x \cos(\theta) + \Delta_y \sin(\theta))}{1 - \frac{2}{r}(\Delta_x \cos(\theta) + \Delta_y \sin(\theta))}.$$

Next, note that the term

$$\Delta_x \cos(\theta) + \Delta_y \sin(\theta) = \left(\sqrt{\Delta_x^2 + \Delta_y^2} \right) \cos(\theta_p - \theta)$$

in polar coordinates represents the radial displacement multiplied by a cosine function, where θ_p is the polar angle of the plasma displacement. Under the assumption of small radial displacement, first-order Taylor expansion is applied to the denominator:

$$S_\theta = \frac{1}{r^2} \times [r - (\Delta_x \cos(\theta) + \Delta_y \sin(\theta))] \left[1 + \frac{2}{r}(\Delta_x \cos(\theta) + \Delta_y \sin(\theta)) \right].$$

By expanding the product and retaining only first-order terms, the approximate form of the magnetic signal is obtained:

$$S_\theta = \frac{1}{r} \times \left[1 + \frac{1}{r}(\Delta_x \cos(\theta) + \Delta_y \sin(\theta)) \right] + \mathcal{O}((\Delta_x, \Delta_y)^2). \quad (\text{A.1})$$

APPENDIX B

CAMERA CALIBRATION

Camera calibration refers to the measurement of camera properties such as location and focal length; complete knowledge of these properties allows projection of any points in 3D space to a 2D pixel plane. Camera properties are divided into two categories known as intrinsic and extrinsic parameters. The intrinsic parameters refer to the camera's internal properties, such as focal length and principal point¹, while extrinsic parameters refer to the camera's translation and orientation in the world coordinate system.

B.1 Homogeneous Coordinate

A simple explanation of homogeneous coordinates [15] is that they are used in computer vision to linearize models and simplify calculations. As an example, consider homogeneous coordinates in two dimensions ($\mathbf{P}^2$). A point on a real 2D plane is expressed as a point in three-dimensional real space excluding the origin, i.e., $\mathbf{P}^2 = \mathbf{R}^3 \setminus \{(0, 0, 0)\}$ in homogeneous coordinates. This is called the 2D projective space. The traditional representation of a 2D coordinate (x, y) in homogeneous coordinates is given by $(\tilde{x}, \tilde{y}, \tilde{z})$, where $\tilde{z}$ is a nonzero real number, with the relations $\tilde{x} = x\tilde{z}$ and $\tilde{y} = y\tilde{z}$. Hence, homogeneous coordinates can be converted to inhomogeneous coordinates by dividing $\tilde{z}$.

To illustrate the benefit of homogeneous coordinates, consider a Euclidean transformation between two coordinate systems in 2D real space from $\mathbf{q}$ to $\mathbf{q}'$, consisting of rotation by a rotation matrix (R) and translation vector ($\mathbf{t}$). The traditional transformation is

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = R \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}. \quad (\text{B.1})$$

¹The point of intersection between the optical axis and the image plane

In homogeneous coordinates, the transformation can be compactly written as

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = [R \quad \mathbf{t}] \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{bmatrix} = [R \quad \mathbf{t}] \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}. \quad (\text{B.2})$$

where $[R \quad \mathbf{t}]$ is known as the extrinsic matrix.

B.2 Point Projection

To project a point in the world coordinate to pixel plane, two transformations are required. The first transformation uses the extrinsic matrix $[R \quad \mathbf{t}]$ to transform the coordinate of a point in 3D world coordinate (u_w, v_w, w_w) to 3D camera-centered coordinate (X_c, Y_c, Z_c) . Using homogeneous coordinates with the camera's rotation R and translation $\mathbf{t}$ in world coordinates, the transformation is performed in $\mathbf{P}^3$:

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = [R \quad \mathbf{t}] \begin{bmatrix} u_w \\ v_w \\ w_w \\ 1 \end{bmatrix}. \quad (\text{B.3})$$

Once the point has been transformed to the camera-centered coordinate system, the intrinsic matrix (K) is used to transform the point in 3D space to 2D pixel space:

$$\begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{bmatrix} = K \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \begin{bmatrix} f_x & s & o_x \\ 0 & f_y & o_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}. \quad (\text{B.4})$$

In 2D pixel coordinates, the axes are x (pixel column) and y (pixel row). The intrinsic matrix holds the following parameters: f_x — linear pixel density along x , f_y — linear pixel density along y , o_x — principal point along x , o_y — principal point along y , and s — sensor axes skew. The homogeneous pixel coordinate is calculated by multiplying K to $\mathbf{q}_c$ (Equation B.4)². Importantly, $\tilde{z}$ in homogeneous pixel coordinates is not arbitrary due to its relation to the world homogeneous coordinate.

Projection of 3D real space to 2D pixel space can be incorporated in one equation by combining Equation B.3 and B.4. This combination results in multiplication of the

²The pixel coordinate is in homogeneous coordinates due to the definition of the intrinsic matrix which requires output in homogeneous coordinates

intrinsic and extrinsic matrix, forming a new matrix called the projection/camera matrix P . Hence the final projection equation is

$$\begin{aligned} \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{bmatrix} &= K \begin{bmatrix} R & \mathbf{t} \end{bmatrix} \begin{bmatrix} u_w \\ v_w \\ w_w \\ 1 \end{bmatrix} = P \begin{bmatrix} u_w \\ v_w \\ w_w \\ 1 \end{bmatrix}, \\ \Rightarrow x = \tilde{x}/\tilde{z} &= \frac{p_{00}u_w + p_{01}v_w + p_{02}w_w + p_{03}}{p_{20}u_w + p_{21}v_w + p_{22}w_w + p_{23}}, \\ \Rightarrow y = \tilde{y}/\tilde{z} &= \frac{p_{10}u_w + p_{11}v_w + p_{12}w_w + p_{13}}{p_{20}u_w + p_{21}v_w + p_{22}w_w + p_{23}}. \end{aligned} \tag{B.5}$$

APPENDIX C

RANSAC

Random Sample Consensus (RANSAC) [15, 16] is an algorithm designed to remove the effect of outliers from model fitting. By applying a penalty score to fitting, outliers can be effectively eliminated. The RANSAC algorithm can be broken down into the following steps:

1. Calculate fitting parameters $\mathbf{p}$ from a randomly selected subset of data points of size k , where k is less than the size of the full dataset.
2. Calculate the residual $\mathbf{r}_i$ of the estimated function $\tilde{x}_i'(x_i; \mathbf{p})$ and the data point $\hat{x}_i'$:

$$\mathbf{r}_i = \tilde{x}_i'(x_i; \mathbf{p}) - \hat{x}_i'.$$

3. Count the number of inliers whose residuals satisfy $\|\mathbf{r}_i\| \leq \epsilon$, where ϵ is a predefined threshold.
4. Repeat steps 1–3 for S iterations.
5. Select the fitted model with the highest number of inliers as the final solution.

To ensure the success of RANSAC, the appropriate number of iterations S is given by

$$S = \frac{\log 1 - P}{\log 1 - p^k}. \quad (\text{C.1})$$

where P is the desired probability of success over S trials, and p is the probability that any given data point is an inlier. For this project, $P = 0.99$ is used with $k = 3$ as the minimum number of points required for parabola fitting, and p is assumed to be 0.5. Therefore, approximately 30 iterations of RANSAC are preferred. Furthermore, it is worth mentioning that several variants of RANSAC exist to increase the algorithm's robustness, such as "Preemptive RANSAC", "PROSAC", and "MLESAC".

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